Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Tunable optical materials for quantum technology*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), **Quantum (√)**			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		*Mechanical and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Optical Materials*	*Surface Treatment*	*Thin-Film Fabrication Technologies*
Concept of Proposed Technology	*Nonlinear optical materials are key to scaling quantum technologies. From photonic integrated circuits for atom and ion-trap systems, switches for photonic quantum computing, entangled light sources, or interconnects for superconducting qubits – new materials are required that outperform the state-of-the-art. In particular, piezo-electric, magneto-optic, and electro-optically active thin films are desired which operate in the cryogenic environments needed for these quantum applications.* *Such thin films can be patterned into quantum photonic devices or heterogeneously integrated with other components to provide quantum functionality. Simply put, without these new materials, the scaling of quantum computing technology is thwarted. Emerging candidates in modulator technology for AI accelerators and classical photonics include Barium Titanate, while Lithium Niobate remains the gold-standard. However, neither of these materials is optimal at cryogenic temperatures.* *We aim to discover, optimize, and harness new tunable materials to scale quantum technology. We have already discovered that Strontium Titanate (STO) is the world's best nonlinear optical and piezoelectric material for quantum applications (1). We also established STO's promise as an element in superconducting qubit applications (2). However, all demonstrations to-date are with bulk crystals.*			

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Rationale for Joint R&D Project	*Our research group can pattern, etch, and transfer oxide thin films, garnets, semiconductors. We have custom cryogenic photonic probe stations for low temperature optical and electrical measurements. We design photonic integrated circuits for quantum and measure qubits in our lab. We also have connections to major quantum computing companies (Google, PsiQuantum, IonQ) with an interest in this area.* ***However, we DO NOT have the ability to grow these materials or process them efficiently.*** *We therefore desire a collaboration that supplies thin films of oxide and other materials for validation in quantum photonic and mechanical devices. Without joint R&D, we cannot study scalable thin film technology, and our partner will not be engaged with the new materials and cutting-edge applications pioneered by our group. We aim to jointly develop active optical, magnetic and mechanical materials, with a specific eye to quantum technologies. A feedback loop is needed between growth and quantum measurement. We would also be interested in collaboration that guides the use cases of novel functional materials for use in quantum technology. Devices produced at UIUC could be externally tested for particular applications.*
Keywords	1. Quantum photonics 2. Photonic integrated circuits 3. Nonlinear optics 4. Thin film devices 5. Materials for quantum information science

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

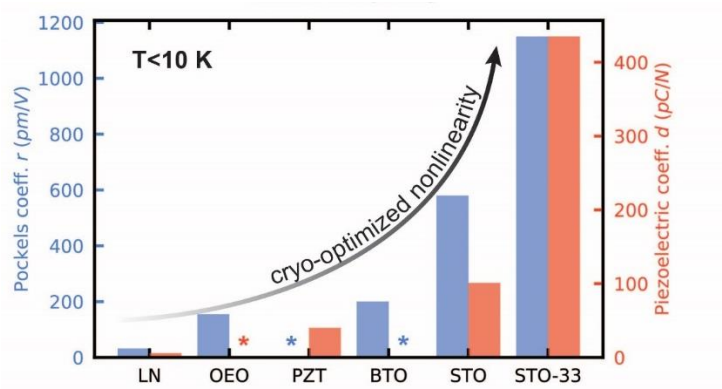

Figure 1: Cryogenic optimized electro-optic and piezo-electric nonlinearity. The state of the art is exceeded by orders of magnitude with new materials (1).

Our general R&D objectives focus on:

- ***Creation of microwave-to-optical transducers.*** *With next-generation nonlinear optical materials studied in our group, the scaling of quantum data centers becomes feasible, with an over 1000x improvement over the state of the art. Our current efforts focus on measurement of nonlinearity of oxide bulk crystals, thin films, and garnet-based optical devices. Integration with silicon nitride photonics and fabrication of superconducting electrodes is our primary workflow. Unfortunately, STO thin films produced by MBE, PLD, and off-axis sputtering still require much optimization (cracking, loss, low performance). Large efforts around the world today focus on BTO, but STO requires further development. We currently work on this project with Google Quantum AI.*

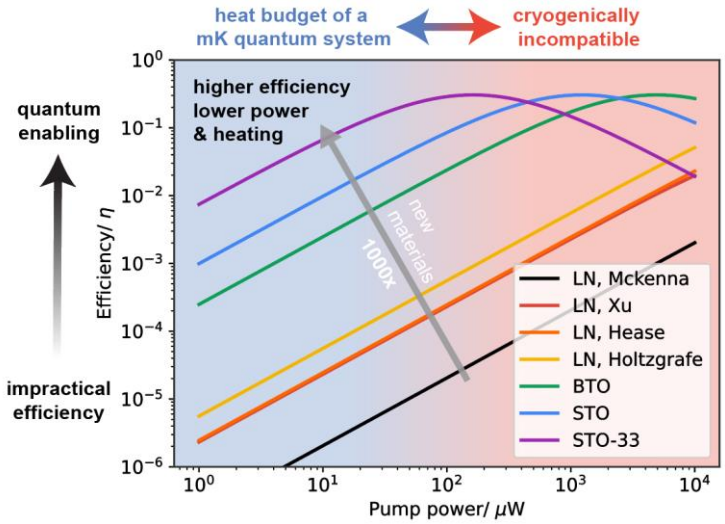

Figure 2: New materials for quantum transduction. STO provides a 1000x boost over lithium niobate circuitry, unlocking modular quantum computing.

- ***Cryogenic optical switches.*** *These same devices revolutionize the size, loss, and power requirements for photonic switching. We aim to demonstrate the lowest VpiL optical modulators at low temperature.*
- ***Optical interconnects.*** *In addition, related systems can constitute an optical interconnect for*

cryogenic systems, reducing the wiring density and thermal budget of coaxial cables. This idea remains a promising solution to scaling large scale quantum systems, but new materials are required to be competitive with the coaxial approach.

- ***Superconducting signal amplification.*** *We aim to use such thin films as tunable varactors (2) that can be harnessed as three-wave mixing elements and amplifiers for cryogenic electronics.*
- ***Next generation cryocoolers.*** *Optimized ferroelectric thin films for cryogenic operation may also function as electrocaloric cryocoolers that replace He3 based systems. Our group works to develop such technology.*

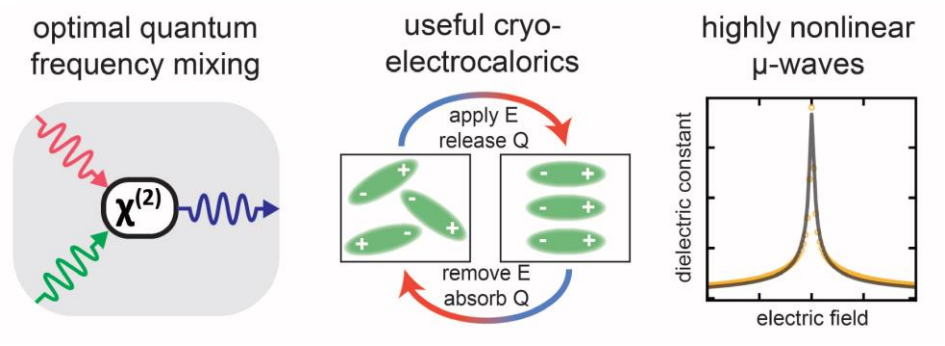

Figure 3: Some applications of highly nonlinear materials designed for quantum applications. Optical frequency mixing and modulation, new cryostat technology, and cryogenic varactors.

Overall, we aim to scale and optimize the STO platform (with doping, strain), discover new and related magnetic materials with strong magneto-optic effects, and prototype devices that chart a path to impact quantum technology. This effort is therefore to develop new materials at sufficient quality and scale to be adopted by large quantum companies. Our current contacts interested in this area include: Google, IBM, Amazon, IonQ, Photonic Inc, Applied Materials, NTT, IMEC, and PsiQuantum.

R&D Target Specifications

Through a potential partnership, we hope to:

- *Increase the scale of thin films of functional optical and mechanical materials beyond what is possible in our lab. We current produce small scale films with a "bond & polish" approach at the mm-scale. Etching technology and processing for films is also of interest.*

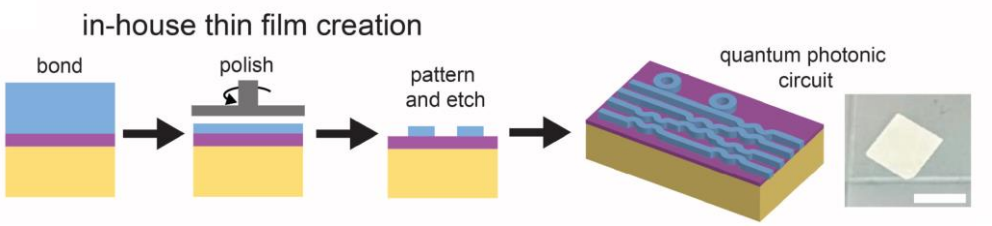

Figure 4: Our current process flow of small-scale quantum photon circuit fabrication. Our lab has expertise in wafer bonding, lapping, polishing, and photonics fabrication.

- *Traditional photonics fabrication, i.e. Silicon Photonics and SiN with an interest in translating quantum technology by adding quantum or active control is of interest. Our research group is not an expert in Si or SiN photonics.*
- *Optical packaging. Many of the applications above require low insertion loss optical links to fiber that survive cryogenic environments. This in itself is an outstanding and worthwhile collaborative effort. (<3 dB at 4K)*
- *Increase the materials flexibility of films of STO. For example. Controlling doping, strain to improve nonlinear response. Superconducting film deposition is also of interest.*
- *Become introduced to emerging quantum applications beyond those mentioned above. In addition, to translate the materials for quantum discoveries to non-quantum applications.*
- *Generally, optical films should be at the 100 mm wafer scale, display losses <1 dB/cm, low temperature dielectric constants >5,000, and Pockel's coefficients >100 pm/V.*

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R&D Plan	*This is difficult to provide without knowledge of the SME partner.* *For Phase 1, we envision that materials transfer and development will occur. Performance and prototype devices will be validated compared to the state of the art. A report on the prospects and performance will be delivered.* *In Phase 2, we envision that a feedback loop is created between quantum device fabrication and measurement (UIUC) and materials development/scaling (Korean SME). By the end of the project, wafer scale films of active optical materials (such as STO) will be integrated into a useful quantum prototype that exceeds the state of the art. Patents will be filed and demonstration collaborative publications submitted. We expect adoption by quantum companies.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*UIUC will fabricate, measure, and characterize thin film materials. Device design and cryogenic measurement will be performed. This will include use of UIUC's nanofabrication facilities and PI Anderson's custom cryogenic and electronic photonic probe stations. Work will be performed by graduate researchers and postdoctoral scholars. UIUC will provide guidance to Korean partners on relevant materials and applications for quantum technologies. UIUC will publish collaborative research articles.*
Expected R&R of Korean SME Partner	*The following is flexible, given the interests and capabilities of the SME partner.* *We expect the SME provides materials growth, processing, transfer and initial characterization of those materials. Materials will be supplied to UIUC as part of the R&D project, not as a paid product.* *We expect the research scope to be primarily in optimizing processes to result in optimal quantum performance.* *The SME may also not focus on synthesis, and instead guide UIUC researchers on applications and prototypes using existing materials platforms (i.e. switches, transducers, modulators). Other topics related to the R&D Target Specifications are of interest.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Korean SMEs will develop materials and products outside of their normal scope in collaboration with UIUC to capture quantum photonics and quantum technology markets. New science will be enabled, leading to high impact publications. Prototypes and patents will be developed. Commercialization will be explored with UIUC's quantum partners and Korean companies. Any quantum industry that requires cryogenic manipulation of microwaves, sound or light will be impacted. Korean companies will become key to the emerging high-tech materials for quantum, while research at UIUC will be accelerated by new capabilities. Specifically, large scale production of optimized nonlinear quantum devices will be established.*

6. References

(1) "Quantum critical electro-optic and piezo-electric nonlinearities". C. P. Anderson, G. Scuri, A. Chan, S. Eun, A. D. White, G. H. Ahn, C. Jilly, A. Safavi-Naeini, K. V. Gasse, L. Li, and J. Vučković. Science 390, 6771, 394-399 (2025)

(2) "Proposal for a 3-Wave Mixing Element with Quantum Paraelectric Materials". E. I. Rosenthal, C. S. Wang, J. Sloan, G. Scuri, Y. Shi, K. Pezeshki, P. M. Noertoft, J. Vučković, C. P. Anderson. arXiv 2510.16621 (2025)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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